



United International University

School of Science and Engineering

Mid Term Examination; Year 2025; Trimester: Fall

Course: PHY 2105; Title: Physics; Sec: All

Full Marks: 30, Time: 1 Hour 30 Minutes

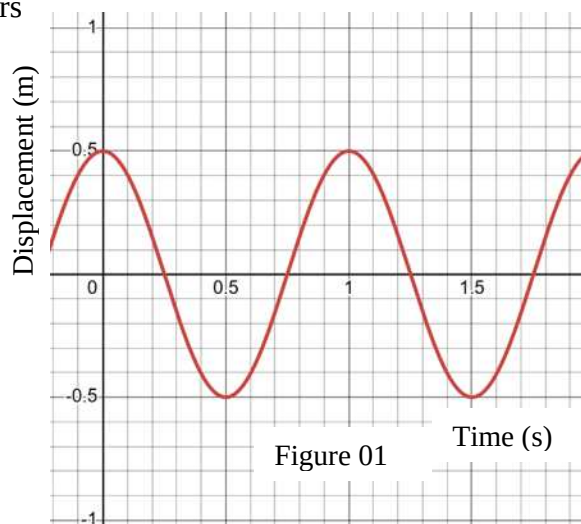
Any examinee found adopting unfair means will be expelled as per UIU disciplinary rules.

Questions no 1, 2, 3, 4 are mandatory to answer. Answer anyone from question no 5 and 6.

- 1.(a) A mass spring system was oscillating in simple harmonic oscillation (SHO) in a closed room. When Ron switched on the ceiling fan the oscillation stopped after a while. Why the oscillation stopped? Draw a diagram representing the displacement vs. time graph for mass spring system after switching on the fan. 2 CO1
- (b) Draw the energy vs. time graph for a mass spring system in simple harmonic motion (SHM) and identify the of maximum potential and kinetic energy in the graph. 2 CO1
- (c) Two progressive waves are propagating in +x axis with same wave speed. wave-1 has larger wave length than wave-2. Which one has maximum frequency. Briefly discuss your logic and draw a diagram of the two waves. 2 CO1

2. A mass spring system oscillates in SHO provided in figure 01. The mass of the mass spring system is 200 g. Calculate the following parameters 6 CO3

- (i) displacement at $t = 0.4$ s
- (ii) acceleration at $t = 0.8$ s
- (iii) potential energy at $t = 0.6$ s



3. The spring constant $k = 100$ N/m, mass $m = 400$ g and damping coefficient $b = 50$ gs^{-1} is found for a mass spring system oscillating in damped harmonic oscillation (DHO). Robert pulled the system up to 20 cm from the equilibrium and released at $t = 0$. 6 CO3

- (i) How many oscillations it will complete in its life time? (life time, $\tau = \frac{2m}{b}$)
- (ii) How much time is required for the amplitude to become 5 cm?
- (iii) What will be the energy of the oscillator after 20 cycle?

4. The equation of progressive wave is 4 CO3

$$y = 5 \sin \pi(100t - 0.003x)$$

Calculate

- (i) wavelength
- (ii) wave velocity

- 5.(a) Derive the differential equation of SHM. Show that $x = A\cos(\omega t + \phi)$ is the solution of the equation. 4 CO2
- (b) Derive the differential equation of DHO for a mass spring system. Write down the conditions for different types of damping. Draw the diagrams for each type. 4 CO2
- 6.(a) Derive the differential equation of DHO in a LRC circuit. Write down the solution for oscillator or underdamped damping considering the current displacement. Draw a diagram of current flow vs. time. 4 CO2
- (b) Show that, the equation of displacement for a progressive wave moving at +x axis is, 4 CO2
 $y = A\sin\frac{2\pi}{\lambda}(vt - x).$

CO1: Define different physical quantities with examples **CO2:** Derive/Show the various equations of SHM, DHM, wave motion etc.

CO3: Evaluate different numerical problems based on the basic characteristics of SHM, DHM.